Design of Enterprise IoT

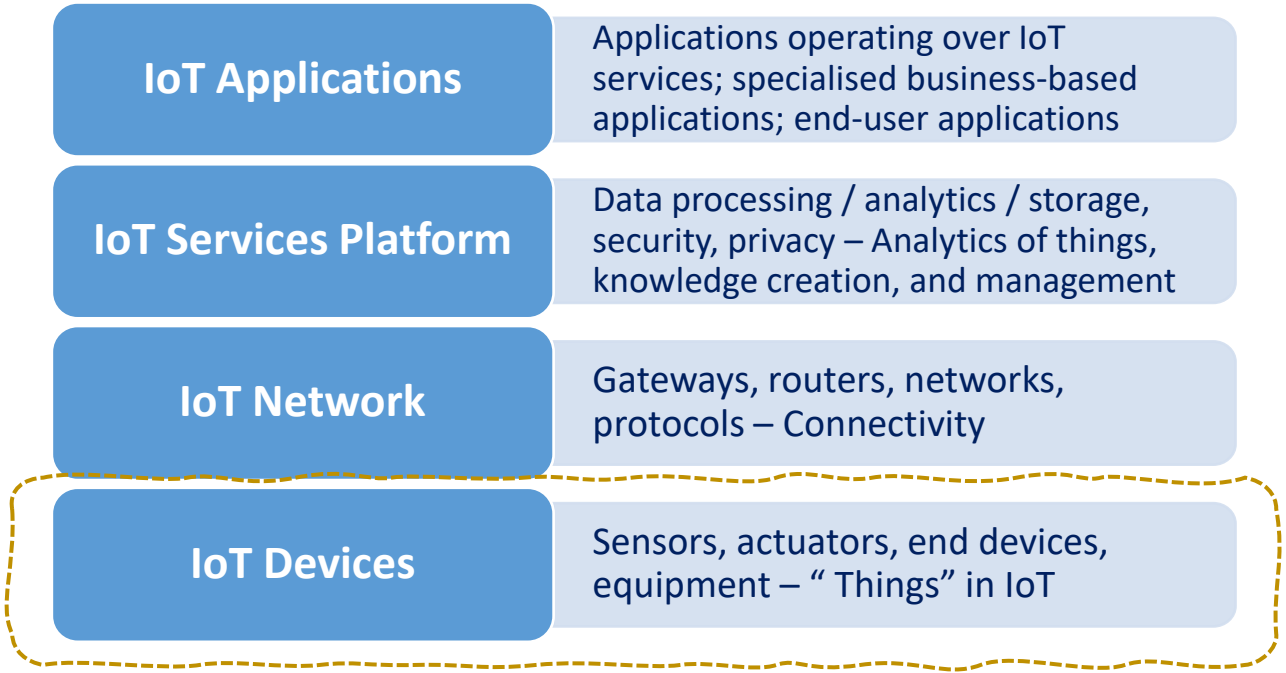


Lecture 2

Sensors & Devices

A/Prof Chun Ouyang

Recap: IoT Reference Framework



(Rayes and Salam, 2022)

“Things” in IoT: Requirements

“Things” are expected to have a digital identity (or a digital representation) and to communicate over the Internet

- **Sensing**: to identify and collect information/data about things for analysis, monitoring, and diagnosis
- **Addressing**: to uniquely identify things over the Internet
- **Actuating**: to control things by performing actions
  - End devices need to provide an interface to application to allow applications to actuate

❖ Sensing and actuating capabilities can be supported on the same device

(Rayes and Salam, 2022)

IoT Devices – Sensors

An electronic device that produces electrical, optical, or digital data derived from a physical condition or event. Data produced from sensors is then electronically transformed, by another device, into information (output) that is useful in decision making done by "intelligent" devices or individuals (people).

(IEEE P1451.6 Terms and Definitions, 2004)

Sensors

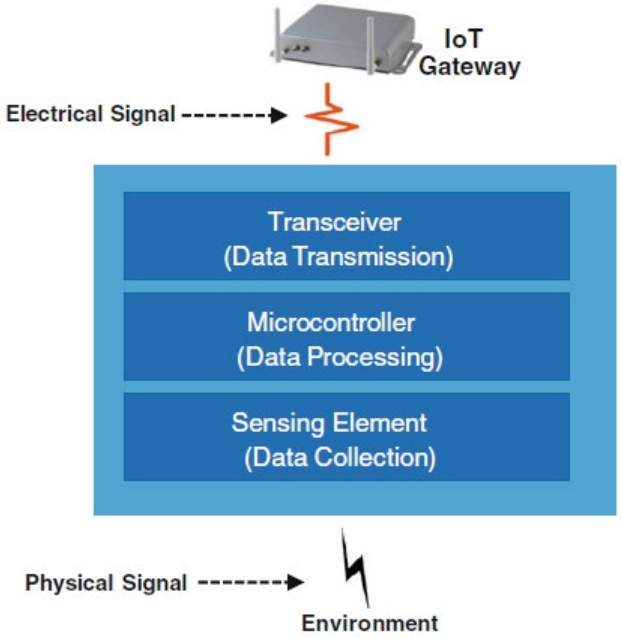
- A device (typically electronic) that detects events or changes in its physical environment and provides a corresponding output
- They transform physical signal into electrical data
  - From analog input to digital output
- They bridge the world's physical objects with the Internet.



(Rayes and Salam, 2022)

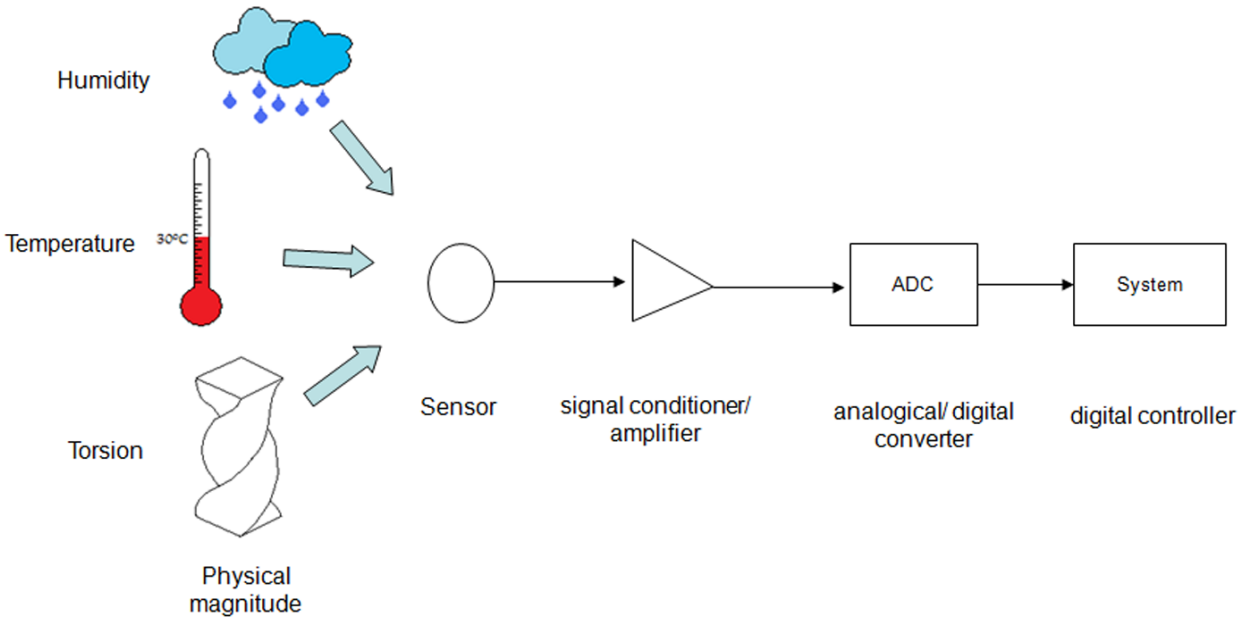
How Sensors Work?

- *Sensing element*: Collect data from its surrounding environment
- *Microcontroller*: Pre-process data and send pre-processed data only
- Transceiver: Transmit output to its adjoining devices or applications



(Rayes and Salam, 2022)

Sensor Components



Sensors and their Application for Disabled and Elderly People (Tomas et al., 2014)

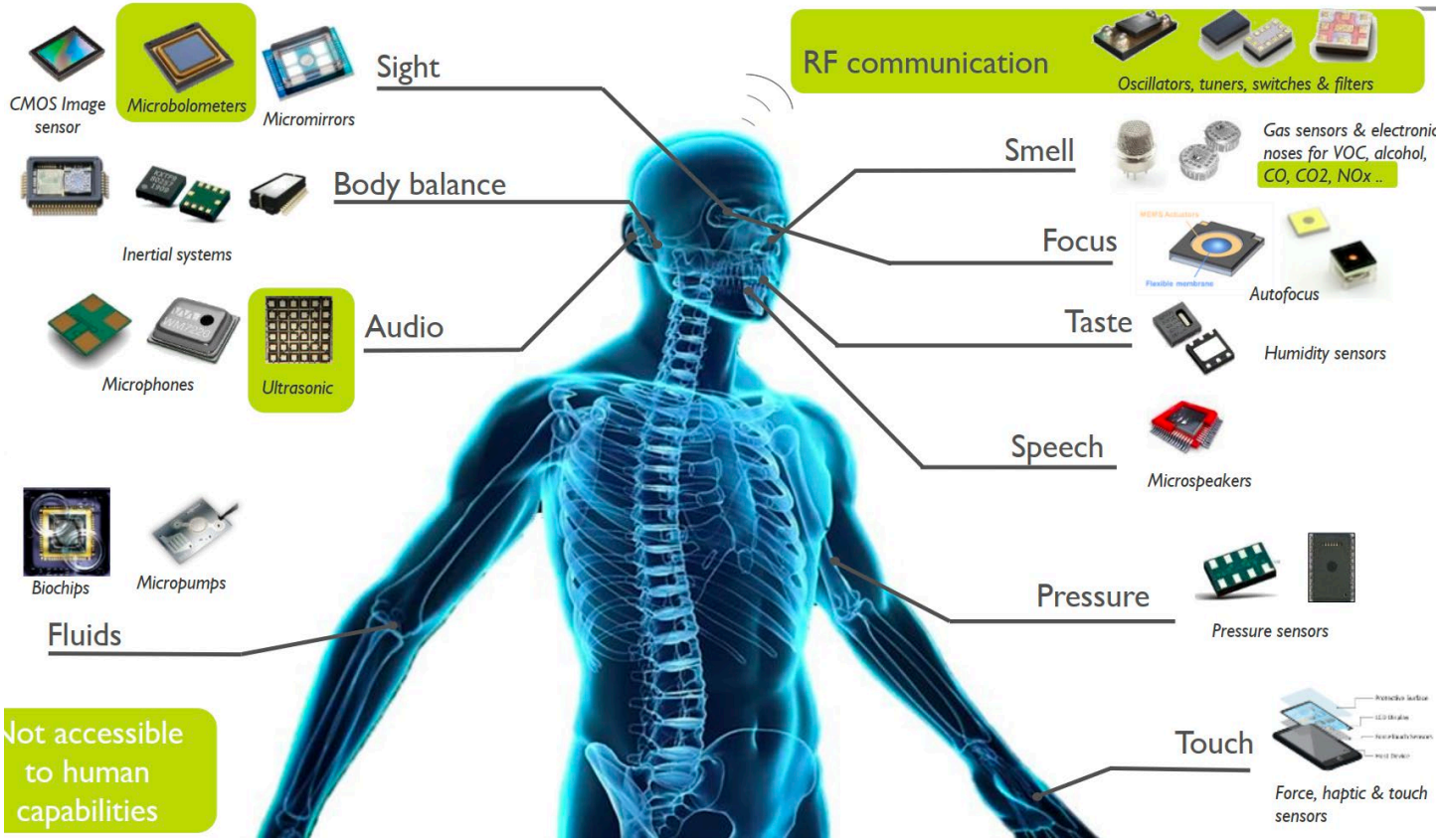
Requirements on Sensors

"Most IoT applications require **smaller** and **smarter** sensors with advanced functionality to collect **more data**, **low-power** processors, **longer** battery life, **faster** response time, and **shorter** time to market."

- Sensors are **Simple and Smart**
  - *Simple*: with a core function to collect and transmit data
  - *Smart*: by providing additional functionality to filter duplicate data, noise, etc.

- Sensors form the front end of the IoT devices. They are comparable to the human five senses
  - *See, Hear, Touch, Smell, Taste*

(Rayes and Salam, 2022)



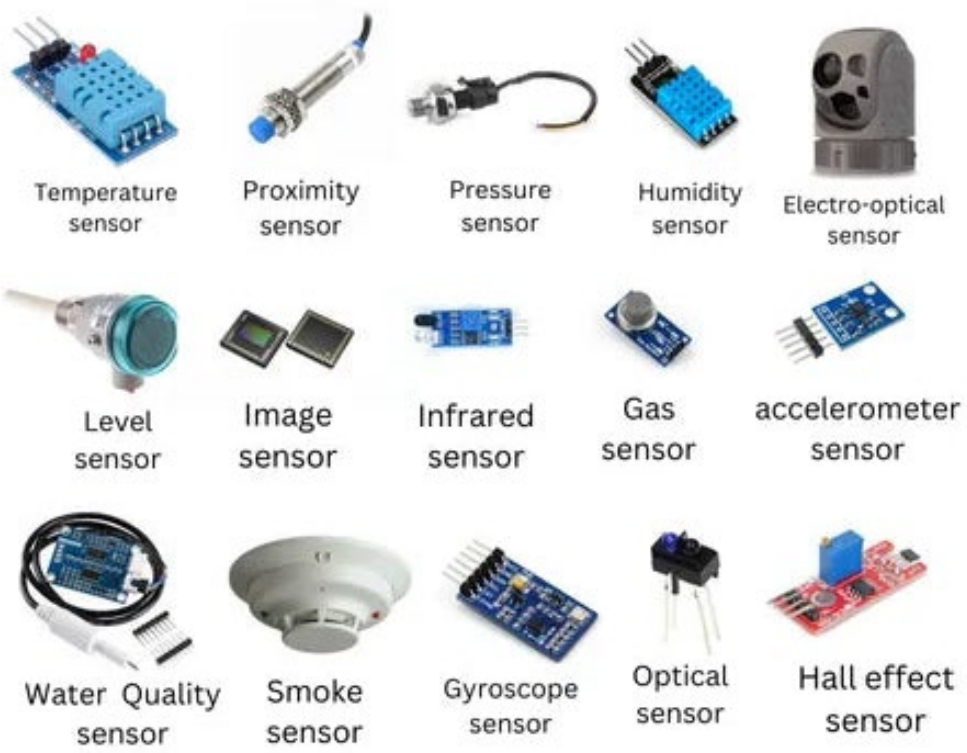
Sensor Types

MEMS & Sensors: beyond human sense

Source: (Yole Developpement)

Sensor Types

- Pressure sensors
  - Pressure of gases or liquid, and nowadays coming with various applications
- Flow sensors
  - Detect and record the rate of fluid flow in a pipe or a system
- Imaging sensors
  - Sophisticated sensors used in digital cameras, medical imaging machines, and night vision equipment.
- Noise sensors
  - Measure noise pollutions
- Air pollution sensors
  - Detect and monitor the presence of air pollution in the surrounding environment
- Moisture and humidity sensors
- Speed sensors
- …



Characteristics of Sensors

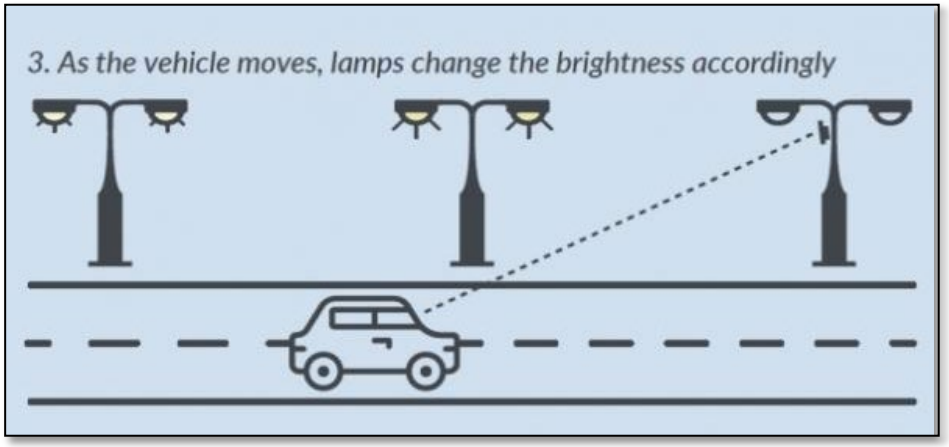
- Data filtering
- Accuracy
- Compact
- Smart detection
- High sensitivity
- Linearity
- Controlled error margin
- Minimum power consumption
- Limited noise
- Wide bandwidth
- High resolution
- Dynamic range
- Higher reliability
- Ease of use
- Minimum interruption

❖ Sensors are only **sensitive** to the measured property: they **don't influence** the measured property.

(Rayes and Salam, 2022)

An Exercise

- Q: What sensors can be used in smart street-lights according to the requirements in the picture?
  - What is to be monitored and controlled?
  - What needs to be measured?
  - What should be “sensed”?
  - …



Other “Sensing” Devices

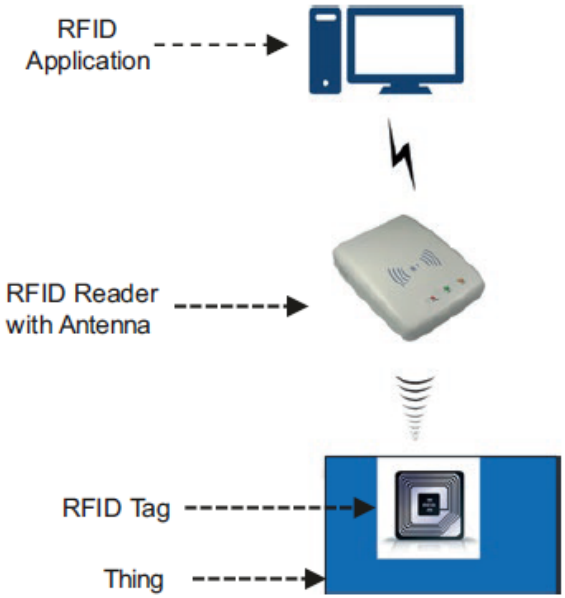
Other ways of capturing information
from “Things” in IoT





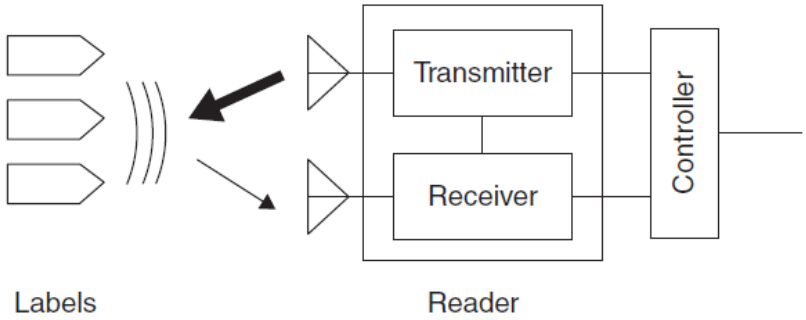
RFID

- RFID stands for Radio-Frequency Identification
- A mechanism to capture information pre-embedded into the “Tag” of a thing using radio waves
- A tag, a reader, and a computer program (i.e. application)



(Rayes and Salam, 2022)

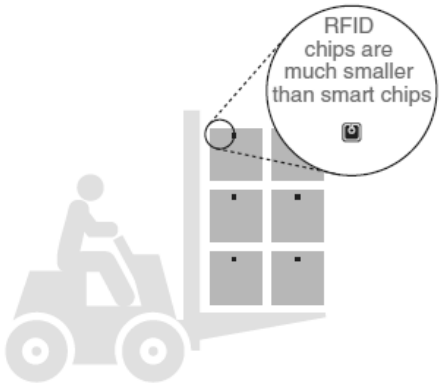
RF Communication



- Step 1: Reader sends out radio frequency electromagnetic waves.
- Step 2: The antenna, tuned to receive these waves, wakes up the chip in the smart card or tag.
- Step 3: A wireless communication channel is set up between the reader and the smart card or tag.

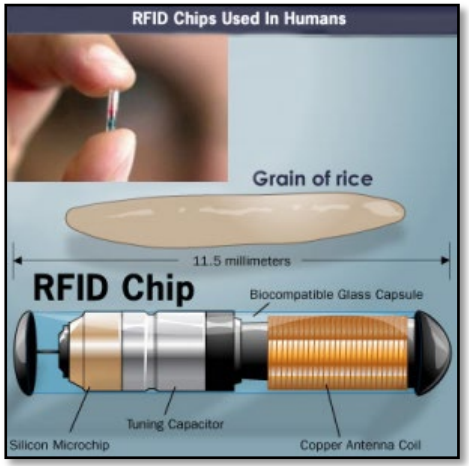
RFID Applications

- Access Control and Management
- Logistics and Supply Chain Tracking
- Transport Management
- Healthcare
- Passport



(Rayes and Salam, 2022)

RFID Injected Wearables





- British scientist Dr Mark Gasson
- RFID nightclub example

Video Tracking

- Process of capturing and analysing the video feeds of an object or a person over a short time interval
- Used for customer identification, surveillance, augmented reality, traffic control, medical imaging, etc.
- Video tracking algorithms & data filtering algorithms
- Pros:
  - Easy to deploy (when sensors/RFID is not available)
  - Comprehensive solution (when working with sensors/RFID)
- Cons:
  - Time-consuming
  - Large amount of video traffic & data to process
  - Complex object/image recognition techniques

(Rayes and Salam, 2022)

Contactless Smart Cards

- Contain a microprocessor that enables
  i. on-board computing,
  ii. two-way communication including encryption, and
  iii. storage of predefined and new information.
- More expensive than RFID



(Minoli 2013, Ch4)

IoT Devices – Actuators

The technological complement to a sensor is an actuator, a device that converts an electrical signal into action.

(https://www.nabto.com/actuators-in-iot-guide/)

Actuators

- Actuators use sensor-collected and analyzed data to monitor and control IoT systems.
- Address and deliver the required performance and action.
  - E.g., shutting down gas flow when the measured pressure is below a certain threshold.
- Transform electrical data into physical actions
  - Digital input to analog output

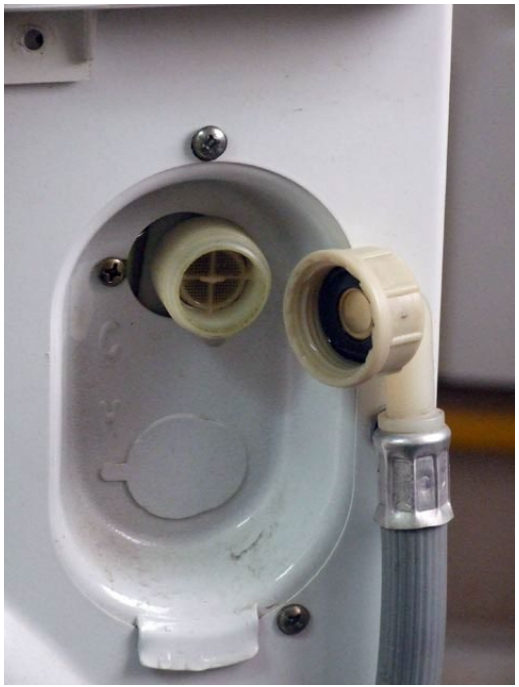


(Source: Mike Vladimer, 2015)

(Rayes and Salam, 2022)

An Example: Washing Machine Water Inlet Valve

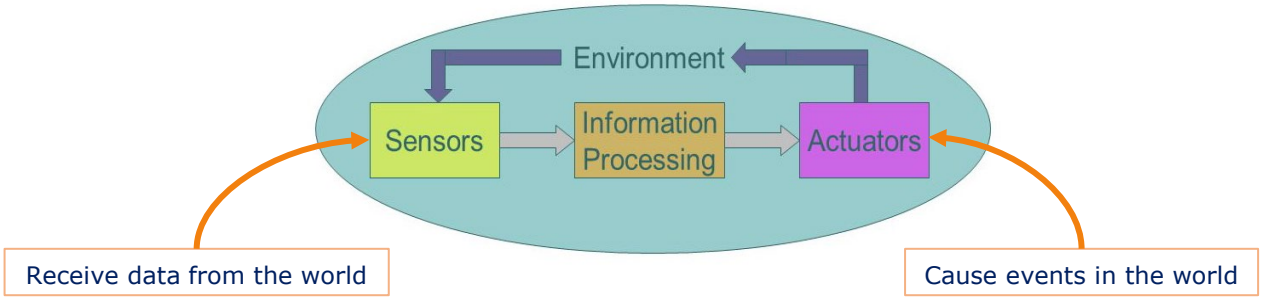
- Press “Start” → the machine doesn't manually turn a faucet
- Inside sits a small **solenoid valve** (electrical actuator)
- Electrical signal arrives → electromagnetic coil pulls a plunger open → water flows in
- Signal stops → spring pushes the plunger back → water stops
- **That's what an actuator does:** turns an “electrical signal” into “physical action” (opening the valve, letting water through)

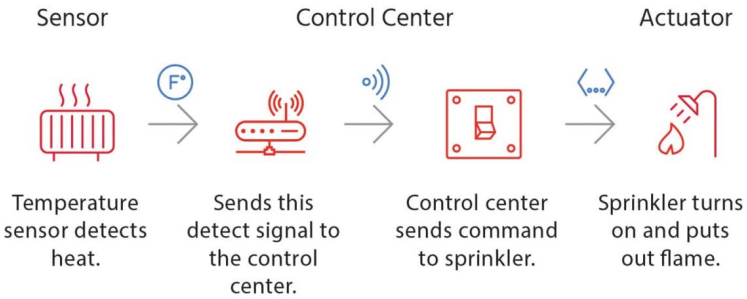


Actuator Types

- Electrical
  - E.g., to control the flow of water into pipes in response to electrical signal
- Mechanical
  - E.g., to convert rotatory motion into linear motion to execute some movements
- Hydraulic
  - E.g. to use pressurized fluid to drive rotary or linear motion with high force
- Pneumatic
  - to use compressed air to drive fast linear motion for clamping or pick-and-place
- Manual
  - E.g., to use human force to move a valve or mechanism through a handwheel, crank, or lever

Interplay of Sensors and Actuators





An example of sensor-to-actuator flow

https://bridgera.com/iot-system-sensors-actuators/

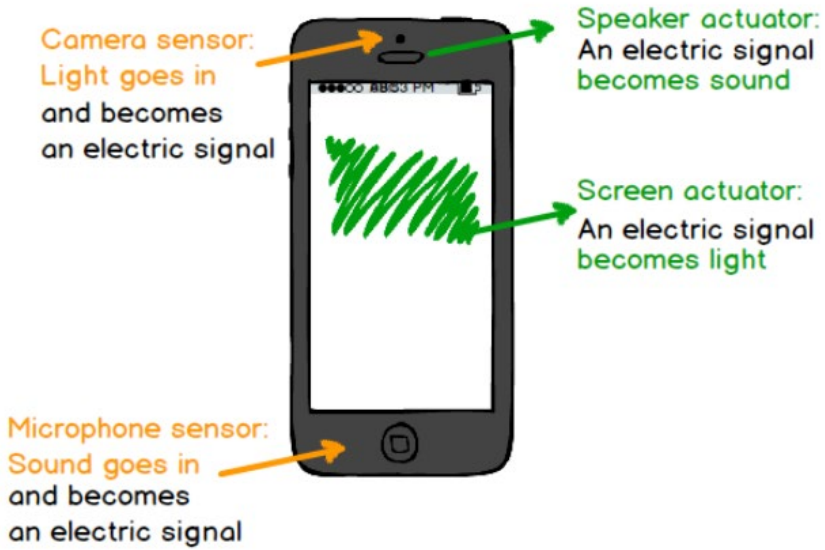
An Exercise

- What are sensors and actuators in a smart phone?



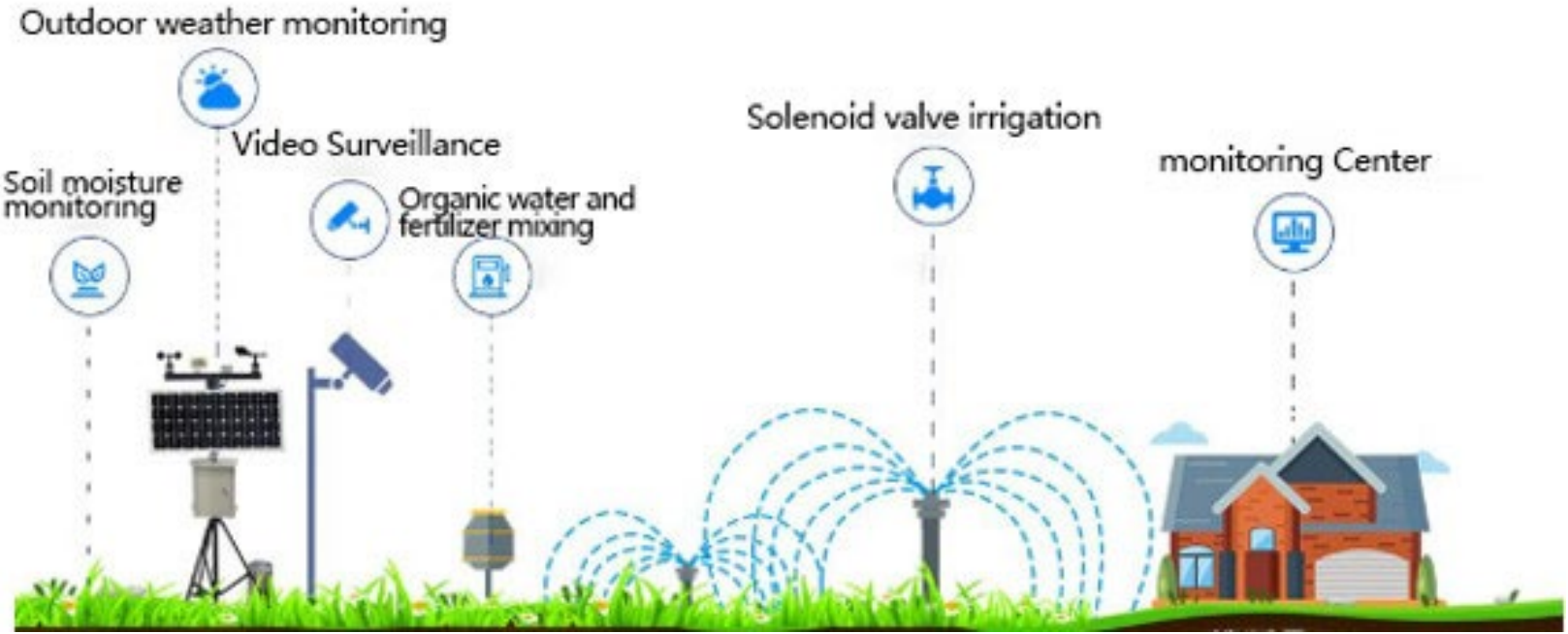
An Exercise

- What are sensors and actuators in a smart phone?



(Source: Mike Vladimer, 2015)

An Example: Soil Monitoring & Irrigation



Smart Toll System (STS)



- *Purpose:* To solve toll collection inefficiency and control traffic by enhancing and speeding up toll collection on freeways, highways, etc.
- *Working mechanism:*
  - Registered vehicles have their unique identities of STS (e-TAG)
  - An enforcement module (or "tag reader") recognizes the vehicle and deducts credits from linked account
  - RFID sensors, laser sensors, light sensors, cameras
- *Data processing*: Information of passing vehicles such as license plate number, section of the freeway, vehicle model, and the time the vehicle passed through